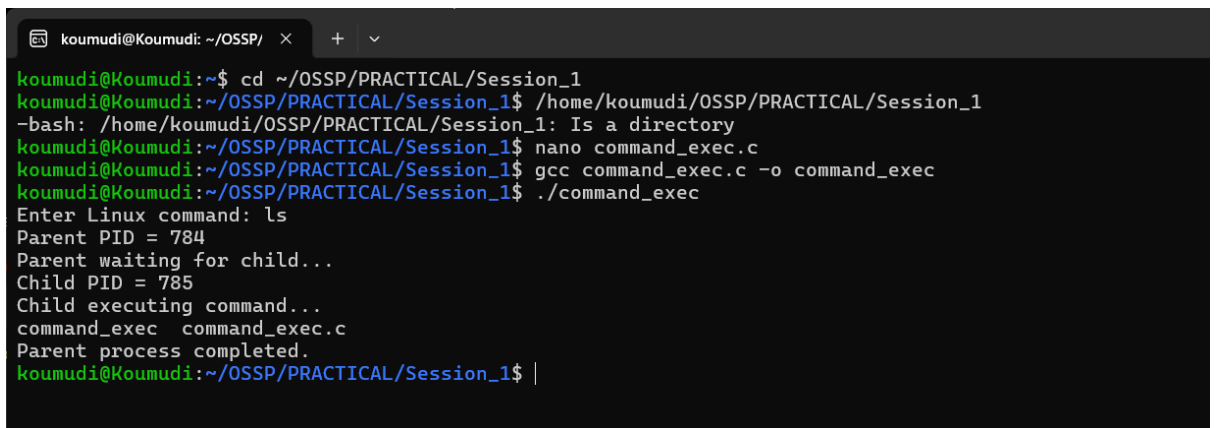


PRACTICAL (OSSP)

SESSION – 1

1. Develop a C program that:

1. Accepts a Linux command as input.
2. Creates a child process using `fork()`.
3. Executes the command in the child using an appropriate `exec()` system call.
4. Allows the parent process to wait for the child using `wait()`.
5. Displays the **Process ID (PID)** of the parent and child processes.



```
koumudi@Koumudi: ~/OSSP/ x + v
koumudi@Koumudi:~$ cd ~/OSSP/PRACTICAL/Session_1
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_1$ /home/koumudi/OSSP/PRACTICAL/Session_1
-bash: /home/koumudi/OSSP/PRACTICAL/Session_1: Is a directory
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_1$ nano command_exec.c
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_1$ gcc command_exec.c -o command_exec
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_1$ ./command_exec
Enter Linux command: ls
Parent PID = 784
Parent waiting for child...
Child PID = 785
Child executing command...
command_exec  command_exec.c
Parent process completed.
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_1$ |
```

2. Using Linux terminal commands: `uname` ,`lscpu`,`lsblk` ,`ps` ,`top`

Investigate the relationship between **hardware resources and operating system services** and prepare a report explaining how the OS abstracts CPU, memory, storage and I/O devices.

```
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_1$ uname -a
Linux Koumudi 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 x86_64 x86_64
GNU/Linux
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_1$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 12
On-line CPU(s) list:   0-11
Vendor ID:              GenuineIntel
Model name:             13th Gen Intel(R) Core(TM) i5-1334U
CPU family:             6
Model:                 186
Thread(s) per core:    2
Core(s) per socket:    6
Socket(s):              1
Stepping:               3
BogoMIPS:               4991.99
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr ss
e sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology tsc_reliable
nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid sse4_1 sse4_2 x2api
c movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowpre
fetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi
1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv
1 xsave avx_vnni vnni umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_cl
ear serialize ibt flush_l1d arch_capabilities
```

```
koumudi@Koumudi: ~/OSSP/ × + ~
Vmscape: Not affected
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_1$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda 8:0 0 356.9M 1 disk
sdb 8:16 0 159.4M 1 disk
sdc 8:32 0 2G 0 disk [SWAP]
sdd 8:48 0 1T 0 disk /mnt/wslg/distro
/
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_1$ ps
PID TTY TIME CMD
549 pts/0 00:00:00 bash
851 pts/0 00:00:00 ps
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_1$ top
top - 16:19:22 up 24 min, 1 user, load average: 0.04, 0.02, 0.00
Tasks: 21 total, 1 running, 20 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7784.6 total, 7200.4 free, 515.0 used, 223.7 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7269.6 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM     TIME+ COMMAND
    1 root        20   0   21876  13148  9668  S   0.0   0.2   0:01.81 systemd
    2 root        20   0   3180   2212  2072  S   0.0   0.0   0:00.04 init-systemd(Ub
    6 root        20   0   3180   2044   1940  S   0.0   0.0   0:00.00 init
   45 root        19  -1   66828  19528  18420  S   0.0   0.2   0:00.54 systemd-journal
   92 root        20   0   25020   6404   5180  S   0.0   0.1   0:01.06 systemd-udevd
  110 systemd+    20   0   21344  13236  11112  S   0.0   0.2   0:00.42 systemd-resolve
  111 systemd+    20   0   91036   7964   6996  S   0.0   0.1   0:00.23 systemd-timesyn
  171 root        20   0    4244   2688   2432  S   0.0   0.0   0:00.03 cron
  172 message+    20   0    9648   5316   4608  S   0.0   0.1   0:00.18 dbus-daemon
```

SESSION – 2

1. Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.


```

koumudi@Koumudi: ~/OSSP/ × + ∨
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MESSAGES", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MESSAGES", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFDIR|0755, st_size=4096, ...}) = 0
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MESSAGES/SYS_LC_MESSAGES", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=48, ...}) = 0
mmap(NULL, 48, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7e087db86000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MONETARY", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MONETARY", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=270, ...}) = 0
mmap(NULL, 270, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7e087db85000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_COLLATE", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_COLLATE", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=1406, ...}) = 0
mmap(NULL, 1406, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7e087db84000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_TIME", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_TIME", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=3360, ...}) = 0
mmap(NULL, 3360, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7e087db83000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_NUMERIC", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_NUMERIC", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=50, ...}) = 0
mmap(NULL, 50, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7e087db82000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_CTYPE", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_CTYPE", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=360460, ...}) = 0
mmap(NULL, 360460, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7e087db29000
close(3) = 0
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}) = 0
openat(AT_FDCWD, "sample.txt", O_RDONLY) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=34, ...}) = 0
fadvise64(3, 0, 0, POSIX_FADV_SEQUENTIAL) = 0
mmap(NULL, 139264, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7e087db07000
read(3, "This is a sample file for strace"... , 131072) = 34
write(1, "This is a sample file for strace"... , 34This is a sample file for strace.
) = 34
read(3, "", 131072) = 0
munmap(0x7e087db07000, 139264) = 0
close(3) = 0
close(1) = 0
close(2) = 0
exit_group(0) = ?
+++ exited with 0 +++

```

SESSION – 3

1. Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

```

+++ exited with 0 +++
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_2$ cd ~/OSSP/PRACTICAL/Session_3
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_3$ pwd
/home/koumudi/OSSP/PRACTICAL/Session_3
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_3$ nano process_states.c
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_3$ gcc process_states.c -o process_states
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_3$ ./process_states
Parent process started.
Parent PID = 1486
Parent PPID = 549

--- PARENT PROCESS ---
Parent PID = 1486
Parent waiting for child...

--- CHILD PROCESS ---
Child PID = 1487
Child PPID = 1486
Child process is running.
Child process terminating.
Parent process continuing after child termination.
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_3$ |

```

2. Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as ps, top, and /proc. Document the observations.

```
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_3$ nano state_demo.c
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_3$ gcc state_demo.c -o state_demo
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_3$ ./state_demo &
[1] 1587
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_3$ Process started. PID = 1587
Process is running...
ps -o pid,ppid,stat,cmd -p 1587
  PID    PPID  STAT  CMD
  1587    549   S     ./state_demo
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_3$ |
```

```
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_3$ Process started. PID = 1614
Process is running...
cat /proc/1614/status | grep -E "Name|State|cat /proc/1614/status | grep -E "Name|State|Pid|PPid"
Name:    state_demo
State:   S (sleeping)
Pid:     1614
PPid:    549
TracerPid:      0
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_3$ Process terminating...
```

```
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_3$ Process terminating...
top
top - 17:17:18 up 1:18, 1 user, load average: 0.08, 0.04, 0.00
Tasks: 22 total, 1 running, 21 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7784.6 total, 7063.4 free, 516.1 used, 360.9 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7268.5 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	21876	13156	9668	S	0.0	0.2	0:02.02	systemd
2	root	20	0	3180	2212	2072	S	0.0	0.0	0:00.04	init-systemd(Ub
6	root	20	0	3180	2044	1940	S	0.0	0.0	0:00.00	init
45	root	19	-1	66828	19628	18512	S	0.0	0.2	0:00.72	systemd-journal
92	root	20	0	25020	6404	5180	S	0.0	0.1	0:01.54	systemd-udev
110	systemd+	20	0	21344	13236	11112	S	0.0	0.2	0:00.45	systemd-resolve
111	systemd+	20	0	91036	7964	6996	S	0.0	0.1	0:00.35	systemd-timesyn
171	root	20	0	4244	2688	2432	S	0.0	0.0	0:00.05	cron
172	message+	20	0	9648	5340	4608	S	0.0	0.1	0:00.29	dbus-daemon
185	root	20	0	17980	8840	7800	S	0.0	0.1	0:00.36	systemd-logind
208	syslog	20	0	222516	5956	4576	S	0.0	0.1	0:00.32	rsyslogd
215	root	20	0	3124	1972	1836	S	0.0	0.0	0:00.01	agetty
222	root	20	0	107036	23016	13560	S	0.0	0.3	0:00.51	unattended-upgr
343	root	20	0	6704	4652	3872	S	0.0	0.1	0:00.02	login
391	koumudi	20	0	20320	11572	9452	S	0.0	0.1	0:00.24	systemd
392	koumudi	20	0	21160	3592	1868	S	0.0	0.0	0:00.00	(sd-pam)
411	koumudi	20	0	6080	5340	3672	S	0.0	0.1	0:00.09	bash
547	root	20	0	3184	1124	980	S	0.0	0.0	0:00.00	SessionLeader
548	root	20	0	3200	1260	1108	S	0.0	0.0	0:00.11	Relay(549)
549	koumudi	20	0	6596	5948	3700	S	0.0	0.1	0:00.16	bash
1256	polkitd	20	0	308172	8152	7256	S	0.0	0.1	0:00.11	polkitd
1631	koumudi	20	0	9300	5680	3496	R	0.0	0.1	0:00.01	top

```
[1]+  Done                  ./state_demo
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_3$ ps -p 1614
  PID TTY          TIME CMD
  1614 pts/0    00:00 ./state_demo
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_3$ |
```